

Integer-valued rational polynomials

Exact integral domains and denominator-aware arithmetic

Universal quotient divisors, complete residue charts, source-bound rational identities and a natural-index Gamma bridge with nonenumerating interval populations.

189

generic and generated
declarations audited

13

source-bound worked packets

19,350

independent signed-domain
checks

Proved arithmetic on exact integral domains. Population execution, generic JSON/compiler refinement and global power-free density retain their separate obligations.

Source, corpus receipts and reproduction commands accompany the repository snapshot.

Exact denominators connect existing arithmetic

The highest-value continuation of the fixed-divisor work is the distinction between a rational polynomial, its integer numerator, and its actual integer outputs. Polynomial normalization and the Gamma interface already produce exact numerator-denominator pairs. The new arithmetic connects those pairs to universal divisibility and local power-free questions while preserving their domains. The source polynomial is given by ascending exact rational coefficients, normalized as $F(x)/L$ with F in $\mathbb{Z}[X]$ and positive L . Integers and rational strings are accepted; floating approximations are rejected. Removing a coefficient denominator is an algebraic rewrite, not permission to substitute the numerator's arithmetic for the quotient's arithmetic.

For example, $F(x)=x(x-1)$ has universal divisor 2. The quotient $x(x-1)/2$ is integral for every signed integer and has universal divisor 1. For exponent 1 the numerator has a universal prime obstruction, while the quotient has none. Higher binomial coefficients make the distinction much larger: the falling factorial of degree d has fixed divisor $d!$, and dividing by $d!$ yields an integer-valued polynomial with fixed divisor 1. Width 64 requires just 65 consecutive samples for integrality and universal divisibility, rather than a search through $64!$ residues.

There is a second distinction on partial integral domains. The rational value lattice of $(x^2+7)/2$ is $(1/2)\mathbb{Z}$, but its actual integer outputs occur only at odd x and have gcd 4. Confusing these two quantities would discard a real universal squarefree obstruction. The executable packet consequently reports `rational_value_lattice`, `numerator_fixed_divisor`, and `integer_domain_fixed_divisor` separately. Zero and an empty integral domain both have universal divisor 0, but the empty-domain flag distinguishes their different meanings.

A finite criterion on the entire integer line

Let D be the fixed divisor of F . The preceding release proves that D is the gcd of $F(0), \dots, F(d)$, for any verified degree bound d . The new Lean theorem `integral_everywhere_iff` identifies integrality at every signed integer with L dividing D . When that condition holds, the fixed divisor of the integer quotient is D/L . The theorem `quotient_divisors_iff` establishes the full equivalence: q divides D/L exactly when q divides every value $F(x)/L$. This includes $q=0$, the zero polynomial, signed constants, and repeated factors. It does not require separability or a derivative Bezout identity.

The quotient divisor is the right object for local admissibility. For every prime p , a k -free value can avoid divisibility by p^k somewhere precisely when D/L is k -free. This is a statement about the absence of a universal local obstruction. It does not produce a single value avoiding all prime powers simultaneously, and it does not prove a positive density. The existing example $(x^2+2)^2$ remains a reminder: local admissibility can coexist with no squarefree outputs.

The denominator also changes the safe congruence modulus. Under integrality, q divides the quotient exactly when Lq divides the numerator. Thus the quotient's q -divisibility is transported modulo Lq . Reducing inputs modulo q alone is generally wrong. For $x(x-1)/2$ modulo 4, inputs 1 and 5 agree modulo 4 but give quotients 0 and 10, which differ in their divisibility by 4. The generic `quotient_mod_transport` theorem proves the correct enlarged modulus for every polynomial and every q .

The natural-index theorem `quotient_natural_divisors_iff` proves the same universal divisor using only nonnegative inputs. A polynomial's degree-plus-one natural sample window already forces its signed divisibility behavior. This is the formal bridge needed by binomial and fixed-width Gamma products, whose original index domain is natural. The Gamma wrapper retains $n \geq 0$ and rejects intervals crossing into negative indices; the signed polynomial extension is a separately declared arithmetic convenience.

Complete charts on partial integral domains

When F/L is not integral everywhere, its integral input set is exactly the union of residues r modulo L satisfying L divides $F(r)$. Every valid input has the canonical coordinate $x=r+Ln$ with r in the finite root set and signed integer n . The generic `integer_domain_cover` theorem proves both directions and handles negative inputs through Euclidean remainder. This is complete domain coverage, not a list of observed examples.

Each residue admits the polynomial chart $G_r(n)=F(r+Ln)/L$ with integer coefficients. The constant coefficient is integral because r belongs to the root set, and every positive-degree coefficient in the affine expansion contains a factor L . The producer performs the exact substitution and checks coefficient divisibility. Native certificates check the source composition identity F composed with $r+LX$ equals L times G_r . Generic `chart_eval`, `chart_integral`, and `chart_quotient` transport evaluation, integrality, and the actual quotient value. Every accepted chart can therefore be passed to existing integer polynomial arithmetic in its own coordinate.

For $(x^2+1)/2$ the unique chart is $x=1+2n$ and $G(n)=2n^2+2n+1$, with fixed divisor 1. For $(x^2+7)/2$ the same coordinate gives $G(n)=2n^2+2n+4$, with fixed divisor 4. For $x(x+1)/6$ the valid residues are 0,2,3,5 modulo 6, and the four charts together have gcd 1. For $x^2/3$ the sole residue is 0 modulo 3 and the quotient chart is $3n^2$. Its universal divisor is 3: exponent 1 is obstructed, but exponent 2 has no universal prime-square obstruction. A constant $1/2$ has no integral residue at all.

The universal divisor on the integral domain is the gcd of the chart fixed divisors. The generic `familyGcd` and `domain_divisors_iff` theorems prove the full universal-divisor equivalence. Certificates also prove local admissibility or exhibit a prime-power obstruction for every integral input. They retain the condition `IntegralAt` in partial-domain statements. In particular, an empty domain gives a vacuous universal divisibility statement and fails the existential local-admissibility requirement; it is never reported as a successful arithmetic solution.

Source identities, execution, and exact populations

A native packet names the exact rational source polynomial and the integer numerator polynomial. The kernel checks their coefficient identity after multiplying the rational source by L , then checks that rational evaluation equals the cast integer quotient at every integral input. The generic `quotient_rational_value` theorem supplies the division bridge. This prevents certificates from silently proving claims about a different integer polynomial after normalization. Each packet lists its declarations and a SHA-256 of its complete source, while the combined audit checks those declarations against the compiled source hash.

For globally integral inputs the domain population of a closed interval is its length. For partial domains, each residue contributes $\text{floor}((hi-r)/L)-\text{floor}((lo-1-r)/L)$. Summing those

contributions counts every integral input exactly once, including signed endpoints. The implementation works with arbitrarily large integer endpoints and never scans the interval. The interval $[-10^{100}, 10^{100}]$ contains exactly 10^{100} odd inputs for $(x^2+1)/2$. These interval counts are exact tested Python arithmetic; this release does not claim a new Lean proof of the population implementation.

The three JSONL operations are `integer_valued_polynomial`, `gamma_integer_arithmetic`, and `native_integer_valued_certificate`. All use the existing query service and ordinary structured arguments. Rational coefficients are exact strings such as `"-1/2"` in ascending order. The native operation returns a source packet that must be compiled and audited before it can be called `kernel_checked`. Individual freshly generated packets deliberately retain `kernel_checked=false`. The release summary alone becomes checked after matching the combined source hash, declaration counts, and allowed axiom sets.

The structural Gamma wrapper uses the existing exact finite-product normalization for binomial, rising, `gamma_shift`, and `finite_product` expressions. The evidence includes binomial widths 8 and 16 with native quotient certificates and a width-four Gamma shift. Python scaling checks widths 8,16,32,64. The wrapper does not claim analytic continuation through Gamma poles or a Lean proof of the normalization producer. The natural-domain universal divisor theorem is proved; general JSON decoding and symbolic compiler execution remain separate obligations.

Reproduction and independent evidence

Run `make integer-valued-check` on the pinned Lean 4.20 toolchain. The script regenerates all exact packets, compiles the generic module and the combined certificate file, prints their axioms, checks hashes and declaration counts, and runs the focused test suite. The release contains 16 reusable theorem declarations and 173 generated declarations across 13 worked packets. Every audited declaration uses only the standard allowed axioms `propext`, `Classical.choice`, and `Quot.sound`. There are no admitted proofs, `native_decide` certificates, or `Lean.ofReduceBool` dependencies.

Twenty focused tests cover denominator cancellation, partial and empty domains, signed charts, the distinction between the rational lattice and integral-output `gcd`, enormous interval counts, Gamma restrictions, bounded work exhaustion, service dispatch, and source hashes. The independent release census contains 150 rational polynomials. It checks 19,350 signed-domain evaluations, 1,350 complete quotient-divisibility periods, 672 signed chart identities, and 150 interval populations. Its Horner evaluator and full modular enumerations are independent of the producer's polynomial evaluator and chart construction. Neighbour tests retain the previously shipped fixed-divisor, Gamma, power-free, and residue-atlas behavior.

Exact producer budgets are degree 128, 2048-bit rational coefficient components and normalized integer coefficients, and a 4096-bit common denominator. Partial-domain enumeration requires `L` at most the declared `period_limit`, capped at 65,536. The global-integrality route skips that enumeration completely. Native certificates require degree at most 32, partial period at most 4,096, at most 32 charts, and chart coefficients at most 4,096 bits. Exact factorization has its existing declared trial budget. Exhaustion raises `WorkLimit`; it never reports incomplete work as a complete answer.

What this unlocks and what remains

Integer-valued polynomial arithmetic now has a proved semantic connection to the repo's integer polynomial machinery. The highest immediate avenue is transporting existing power equations and repeated-factor routines through these exact charts, including original-coordinate image constraints. A denominator can cancel part of a repeated factor's prime valuation, so the old inference $g(x)=\pm 1$ must be applied to an integral chart with its checked factorization, rather than directly to an arbitrary rational numerator. Chartwise finite solvers can be unioned only after proving complete domain coverage and transporting their returned coordinates.

Efficient high-denominator partial domains are another opportunity. Complete Hensel lifting and CRT can replace enumeration modulo L , but the resulting compressed domain needs a precise chart or semilinear representation and its own source-bound certificate. The present implementation chooses the exact small-period route and explicitly refuses larger partial periods. This limitation does not affect globally integer-valued binomial products, which use the finite-window criterion.

General compiler and JSON refinement, formal Gamma normalization execution, and global density for separable power-free polynomial values remain open. Across the repo, certified Sturm execution, effective unit-equation bounds, and rigorous period enclosures continue as independent work avenues. This release supplies a concrete arithmetic bridge and checked source identities without treating those larger research programs as completed. The complete release archive preserves the rest of the repository and can reproduce this advance alongside the other sessions' work.